Sports Medicine

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Purpose.—This section provides current information related to the medical needs of young athletes, as pertinent to counseling young athletes and their parents regarding sports participation and practices contributing to the health maintenance of the athlete, as well as current concepts in the prevention, diagnosis, and treatment of sports-related illnesses and injuries.

Editorial Comment.—*The prevalence of injuries in sports varies with multiple factors, including the sport, the environment, and the participant. In this prospective study of soccer injuries, the authors are able to epidemiologically identify the factors most frequently encountered in a youth and adolescent soccer program. With knowledge of the type and cause of the injuries, the incidence can be reduced in the future by appropriate training and skill development.*—W.B.S.

Soccer Injuries and Their Relation to Physical Maturity

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• A prospective study of injuries encountered during participation in a summer soccer camp for youths aged 6 through 17 years revealed an injury incidence of 10.6 per 1000 hours for girls (107 injuries in 458 girls) and 7.3 per 1000 hours for boys (109 injuries in 681 boys). For both sexes, the incidence of injury increased at age 14 years. One certified athletic trainer (W.D.C.) assessed and documented all injuries during the course of the study. Seventy percent of injuries involved the lower extremities. Contusions represented 35.2%, strains 27.8%, sprains 19.4%, and fractures and dislocations 2% of all reported injuries. The ankle was the most frequent site of injury in both sexes. Quadriceps strain was a common injury in boys. The boys with the highest incidence of injury were tall (>165 cm) and had a weak grip (<25 kg), suggesting that skeletally mature but muscularly weak boys may be more susceptible to injury while playing soccer with peers of the same chronological age.
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Over the past 15 years, soccer has rapidly increased in popularity among young people, with over 1.88 million players under the age of 19 years in the United States[1] and nearly 60 million young and recreational soccer players worldwide.[2] This widespread participation has been accompanied by a large number of injuries. Soccer injuries are reported to be only one fifth as frequent as and less severe than injuries sustained in American football.[3,4] Physically immature adolescents are more prone to injury when playing football with age-matched peers who are more physically mature.[5] In this group of sports participants, the period of greatest growth velocity has been postulated as the time musculoskeletal injuries are most likely to occur.[6] There has been no documentation that risk of soccer injury is related to maturational status.

The present study prospectively examined the risk of injury within groups according to age, gender, and physical maturity in the controlled environment of a one-week soccer camp.

MATERIALS AND METHODS

Data were collected prospectively during five one-week sessions from all boys and girls attending a summer soccer camp in northwest Washington state. Parents were notified of the scope and purpose of the study by mail before the data collection began. The study sample included 681 boys and 458 girls aged 6 to 17 years. All subjects

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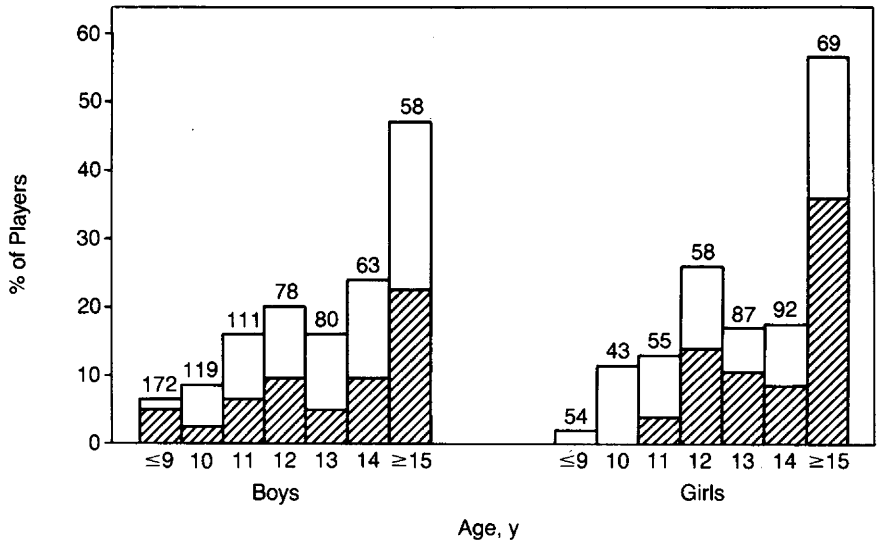
participated in 4½ days of camp activity, averaging a total of 22 hours of soccer. The soccer activity was roughly divided into eight two-hour blocks of group skills instruction and four 90-minute sessions of team competition. Each team was composed of five players with varying skill levels but of the same chronological age. The first week was designed primarily to provide goalkeeper training and advanced field play; the other four camp sessions covered positional play, and skill levels were mixed. Divisions were established by the soccer coaches according to chronological age. During the fifth session, the ratio of girls to boys decreased from 1:2 to 1:3 due to a Washington Interscholastic Activities Association rule prohibiting camp attendance after Aug 1 for girls who intend to play high school soccer at the beginning of the fall school term. All training and competition was on nonirrigated and relatively hard outdoor grass fields that were exposed to precipitation up to three times per week.

Grip strength, weight, and height were measured for all male participants. A grip strength of 25 kg was used to identify the boys as either physically immature (Tanner stages 1 to 3) or mature (Tanner stages 4 and 5); this is similar to the criterion used by Kreipe and Gewanter.[7] Grip strength was measured using a hand-held dynamometer (Jamar, Asimow Engineering Co, Los Angeles). Single trials were obtained for each hand, with informal peer-group competition. The two measurements were then averaged.

Combining height with grip strength further improved the identification of Tanner stage in the boys, with 100% specificity and 89.8% sensitivity (D.D.B. and J. A. Farrow, MD, unpublished findings, 1988). According to this scheme, boys less than 165 cm in height and with an average grip strength below 25 kg are immature (Tanner stages 1 to 3), while boys with greater height and grip strength are very likely to be mature (Tanner stages 4 and 5). Some adolescent males are pubertally mature but do not demonstrate a grip strength of 25 kg or greater. Thus, these boys are tall but weak, possibly indicating greater sexual maturity than muscular development. We examined the injury experience of three groups of boys in the present study: immature, mature, and mature but weak. Girls were classified by chronological age alone, because no similar increase in muscular strength, as measured by grip strength, with sexual maturity has been noted in females.[8]

Injuries were defined as any soccer-related medical problem causing the player to miss full participation in one or more sessions of competition or instruction; a session consisted of a two-hour block of training or competitive play. A single trainer certified by the National Athletic Trainers Association (W.D.C.) observed all injuries during the course of this study and recorded data regarding each injury in a rigidly monitored format. The form categorized when and how an injury occurred, the area injured, the type of injury, the style of shoes and shin guards worn at the time of injury, treatment, and the number of instruction sessions missed. Lower-extremity injuries were further categorized by mapping the areas involved on anterior and posterior below-the-knee diagrams. All report forms were reviewed daily by investigators on the study team.

Data were analyzed using frequencies and cross-tabulation procedures (SPSS Inc, Chicago). Statistical comparisons were made using a test for differences between proportions when comparing two frequencies and using a χ^2 test in the evaluation of injury incidence by age. The incidence of injuries was calculated relative to the number of observed player-hours and/or relative to the number of individuals within categories.



Percentages of boys and girls injured within age groups ($P<.01$ for each gender, χ^2 test). Hatched areas indicate injuries that resulted from contact with another player; open areas, noncontact injuries. Numbers above bars indicate sample sizes in age groups.

RESULTS

Incidence of Injuries

A total of 254 injuries were observed among an overall sample of 681 boys and 458 girls during the five-week data collection period. Thirty-eight injuries occurred during nonsoccer activities and are not included in this report, which considers only the 216 injuries directly related to soccer training and play. The injuries were equally divided between girls (107) and boys (109), indicating a somewhat greater incidence of injury for girls (10.6/1000 hours) than for boys (7.3/1000 hours). In both sexes, the injury incidence within age groups was relatively constant up to age 14 years and then doubled (Figure). The injury incidence was similar during morning (27.8%), afternoon (34.7%), and evening (31.5%) sessions. Participants in the week of specialized and advanced play (first week) had an injury rate of 21.8%, while there was a progressive decrease in the proportion of players who were injured from the first to the last of the four regular weekly sessions (28.7%, 23.2%, 16.7%, 10.0%). There was no apparent relationship between this changing incidence of injury and the total number of participants, the sex ratio, or playing field conditions. There was, however, a corresponding reduction in the number of older girls, who fall in the age category of greatest risk for injury, due to the previously mentioned Washington Interscholastic Activities Association rule.

Type and Mechanism of Injury

Contusions (35.2%), strains (27.8%), and sprains (19.4%) accounted for most of the injuries. Half of all injuries (101) resulted from contact with other players. Sprains and strains were primarily noncontact injuries (74.0%),

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Frequency of Soccer Injuries by Type and Location

Injury Type/Location	No. of Injuries (% of Total)		
	Total	Boys	Girls
Sprain/ankle	25 (11.6)	8 (7.3)	17 (15.9)
Contusion/knee	19 (8.8)	8 (7.3)	11 (10.3)
Strain/leg	19 (8.8)	11 (10.1)	8 (7.5)
Strain/quadriceps	18 (8.3)	12 (11.0)	6 (5.6)
Contusion/leg	15 (6.9)	8 (7.3)	7 (6.5)
Contusion/ankle	12 (5.6)	8 (7.3)	4 (3.7)
Blisters/foot	10 (4.6)	5 (4.6)	5 (4.7)
Contusion/foot	8 (3.7)	4 (3.7)	4 (3.7)
Contusion/nose	8 (3.7)	3 (2.8)	5 (4.7)
Contusion/head	7 (3.2)	5 (4.6)	2 (1.9)
Strain/groin	6 (2.8)	4 (3.7)	2 (1.9)
Strain/back	6 (2.8)	2 (1.8)	4 (3.7)
Sprain/fingers	6 (2.8)	2 (1.8)	4 (3.7)
Strain/knee	4 (1.9)	4 (3.7)	0
Abrasion/knee	4 (1.9)	3 (2.8)	1 (0.9)
Sprain/wrist	4 (1.9)	2 (1.8)	2 (1.8)
Strain/foot	4 (1.9)	2 (1.8)	2 (1.8)
Strain/ankle	4 (1.9)	1 (0.9)	3 (2.8)
Other*	37 (17.1)	17 (15.6)	20 (18.7)
Total	**216** (100.2)†	**109** (99.9)†	**107** (99.8)†

*Includes all injuries that occurred less than four times.
†Percentages do not add up to 100% because of rounding.

while contusions were related to player contact (93.2%). The most severe injuries observed were two knee sprains and two fractures. Both knee sprains were experienced by 15-year-old girls with a history of knee injury. One fracture occurred in the distal fibular epiphyseal plate in the ankle of a 14-year-old girl, and the other occurred in the fifth metacarpal of a 16-year-old boy. Of the injuries, 81.2% resulted in only one missed training session, 12.1% resulted in two missed sessions, and 2.8% resulted in three missed sessions (corresponding to a full day of activity). Only 3.0% of injured players missed more than one day as a result of their injuries.

Lower-Extremity Injuries

Lower-extremity injuries accounted for 153 injuries (70.8%). Only one hamstring injury was observed. Quadriceps strain was the most frequently observed injury in boys and had a significant right-sided predominance (81%). Sprained ankles occurred most frequently in girls (Table).

Failure to wear shin guards markedly increased the proportion of leg injuries. Of all injured players who wore shin guards, only 2.2% (1/45) sustained an injury to a leg. Of all players documented as not wearing shin guards at the time of injury, 10.5% (16/152) had leg injuries. On the basis of shin guard use reported among the injured players, excluding those with leg injuries, 24.4% of players wore some kind of shin guard.

Injuries and Maturity Level in Boys

The injury rate was determined by age and maturity level for the boys in the study. Of 681 boys measured, 7.6% (52) had average grip strengths of 25 kg or more, and all of these boys participated in the same camp division. Those with an average grip strength of at least 25 kg accounted for 48.1% of the 17-year-olds, 29.3% of 15- to 16-year-olds, 7.6% of the 13- to 14-year-olds, and none of the boys 12 years old or younger. The injury rate was not significantly different for strong boys (30.8%) compared with weak boys (21.0%) playing with age-matched groups in the same division.

When the data were analyzed in relation to height and grip strength of the male participants, there was a significantly higher incidence of injuries among the tall and weak boys (24.8%) compared with the immature (short and weak, 13.5%) and mature (tall and strong, 17.3%) boys ($P<.05$). The most frequent injuries in the tall and weak boys were knee contusions (27.3%) and lower-extremity strains (18.2%); in the immature boys, strained quadriceps (23.7%) and sprained ankles (15.8%); and in the mature boys, lower-extremity contusions and abrasions.

COMMENT

Previous studies of soccer injuries among youths include two detailed analyses of insurance claims,[3,9] a telephone survey,[10] a mailed questionnaire,[11] and analyses of data gathered at two European youth soccer tournaments.[12,13] In these studies, two thirds to three fourths of all soccer injuries among youths involved the lower extremities. During intense competition, as in tournaments, lower-extremity contusions were the most frequently observed injury,[12,13] while in other soccer play, ankle sprains were most frequently reported.[3,9,10] The injury experience in the present report is similar to the experience in these earlier studies, with ankle sprains occurring most frequently. However, the present findings demonstrate a high incidence of right-sided quadriceps strains in male participants.

More serious injuries, such as fractures and knee sprains, accounted for only 2% of the injuries in this study, an occurrence of 0.16 serious injuries per 1000 hours of play. Knees are an expected site of serious injury and account for 21.6% of adult soccer injuries; these commonly involve sprains and meniscal injury.[4] Serious knee injuries have been reported to occur in less than 0.1% of all children in competition.[12,13] The proportion of serious injuries in the present study is also lower than the proportion in the samples reported to insurance companies; 9.0% to 13.8% of claims made were for serious injuries.[3,9]

The overall incidence of injury in our prospective study was 10.6 per 1000 hours for girls and 7.3 per 1000

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hours for boys. These values are higher than those reported in off-site surveys, in which minor injuries tend to be underreported,[10,11] and lower than those encountered in more intense tournament competition.[12,13]

The relationship between age and increasing incidence of soccer injury noted in our study has been previously documented.[9,10,13] Several of these studies also noted the abrupt increase in injury incidence near age 14 years. Girls are consistently reported to have a higher incidence of injury than boys, and they show the same trend toward increased injury around age 14 years.

Instead of a clustering of injuries in the immature boys, who are theoretically most susceptible, a higher rate of injury was observed in tall boys who were classified as Tanner stage 4 or 5. An increased incidence of injury was particularly evident in boys who were more than 165 cm tall but had an average grip strength below 25 kg. More aggressive play and greater risk taking associated with maturity may be reflected by the high incidence of contusion injuries in the clearly mature (tall and strong) boys. Lack of coordination and strength in the boys who were mature according to Tanner stage but had a weak grip could also be a factor leading to injury. These weak but pubertally mature boys had a higher incidence of the same types of injuries seen in immature boys. Thus, pubertal maturity does not appear to be an important discriminator of who will be injured playing soccer. Instead, injuries might be more effectively reduced by matching participants according to some measurement of muscular development, since this generally lags behind pubertal and skeletal maturity in the developing adolescent.

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